

MEMORY-CONDITIONED GUIDED PPO FOR DIVERSE UAV TRAJECTORY COLLECTION

Zhiyan Li

ABSTRACT

This report studies diversified trajectory collection for UAV navigation in a 2D urban environment. We focus on improving trajectory diversity while preserving task feasibility, using a hierarchical pipeline that combines dynamic-cost multi-path A*, turning-point landmark pooling (with deduplication and clustering), and memory-penalized goal-conditioned PPO control. On the official evaluation protocol, our method improves overall diversity score from 1996.41 (simple PPO) to 2369.41 (+18.68%), with 15 out of 20 initials improved. We further provide per-initial analysis, training behavior discussion, and qualitative trajectory comparisons.

1 INTRODUCTION

Learning robust vision-language navigation (VLN) policies for unmanned aerial vehicles (UAVs) requires large-scale trajectory data with both high task success and rich behavioral diversity. In practical data collection pipelines, however, agents trained with conventional goal-reaching rewards often converge to a single dominant route for each start-goal pair. Although such trajectories may be efficient, they provide limited coverage of alternative navigation behaviors and therefore constrain the generalization ability of downstream VLN or VLA models.

From a research perspective, this problem is closely related to exploration in continuous spaces, multimodal policy learning, and diversity-aware optimization. Existing single-policy RL training often prioritizes expected return and may under-represent alternative successful modes. Therefore, this task encourages strategies such as reward shaping, conditional policies, memory-based anti-collapse mechanisms, population training, or planning-RL hybridization to jointly optimize success and diversity.

In summary, the goal of this work is to study how to build a trajectory generation pipeline that scales to many start-goal configurations and produces diverse successful demonstrations suitable for downstream policy learning. We treat diversity as a first-class objective alongside goal achievement, and we use the provided benchmark and evaluation tools to quantify this trade-off systematically.

2 METHOD

Our method is a hierarchical pipeline with three stages: (i) dynamic-cost multi-path A* to generate globally distinct feasible routes, (ii) turning-point pooling with deduplication and clustering to form robust landmark candidates, and (iii) memory-penalized landmark selection with goal-conditioned PPO for local control.

2.1 DYNAMIC-COST MULTI-PATH A* PLANNING

For each start-goal pair, we build an occupancy grid from the 2D point cloud and run A* repeatedly with a dynamic cost map:

$$\mathcal{C}_k(r, c) = 1 + \lambda_{\text{rep}} w_{\text{fade}}(r, c) M_k(r, c) + \epsilon_k, \quad (1)$$

$\mathcal{C}_k(r, c)$ is the A* planning cost at cell (r, c) in iteration k , M_k is planning memory, w_{fade} weakens repulsion near goal, and ϵ_k is a noise.

For a grid cell u , we define the repulsion injection as

$$\text{Repulse}(P_k; s_{\text{rep}}, r_{\text{rep}})(u) = s_{\text{rep}} \sum_{v \in P_k} \mathbf{1}(d_{\text{grid}}(u, v) \leq r_{\text{rep}}), \quad (2)$$

For each cell u , repulsion accumulates contributions from path cells v within radius r_{rep} , s_{rep} sets strength and $\kappa(\cdot)$ controls distance decay.

After obtaining path P_k , planning memory is updated as

$$M_{k+1} = \text{clip}(\gamma M_k + \text{Repulse}(P_k; s_{\text{rep}}, r_{\text{rep}}), 0, M_{\text{max}}), \quad (3)$$

M_{k+1} is the updated planning memory map, γ controls decay, $\text{Repulse}(\cdot)$ adds penalties around the current path, and $\text{clip}(\cdot)$ bounds memory into $[0, M_{\text{max}}]$. To avoid unrealistic detours, we keep only paths satisfying

$$\text{Len}(P_k) \leq \rho_{\text{max}} \text{Len}(P_{\text{shortest}}). \quad (4)$$

The accepted path length must not exceed ρ_{max} times the shortest-path length.

2.2 TOP- K SHAPE-DIVERSE PATH SELECTION

From N candidate paths, we compute a turn-shape feature vector

$$\phi(P) = [\text{Len}(P), \sum |\Delta\theta|, \sum \Delta\theta, n_{\text{left}}, n_{\text{right}}, n_{\text{turn}}, d_{\text{turn}}, h_1, \dots, h_6], \quad (5)$$

where $\text{Len}(P)$ is total path length, $\Delta\theta$ is the heading change, n_{left} and n_{right} are the numbers of left/right turns under an angle threshold, and $n_{\text{turn}} = n_{\text{left}} + n_{\text{right}}$ is the total number of effective turns. d_{turn} denotes the mean length between turn points along P . $h_1, \dots, h_6$ are a 6-bin normalized histogram of local turn angles $\Delta\theta$, capturing coarse turn-distribution shape.

We normalize ϕ and apply greedy farthest-point sampling:

$$\mathcal{S}_{m+1} = \mathcal{S}_m \cup \arg \max_{i \notin \mathcal{S}_m} \min_{j \in \mathcal{S}_m} \|\tilde{\phi}(P_i) - \tilde{\phi}(P_j)\|_2, \quad (6)$$

where $\mathcal{S}_m$ is the index set of already selected paths after iteration m . Starting from an initialized seed set, we iteratively add one path until $|\mathcal{S}_m| = K$.

The selected Top- K paths form a compact but shape-diverse subset that is then used as the input of Section 2.3 for turning-point extraction and landmark construction.

2.3 TURNING-POINT EXTRACTION, DEDUPLICATION, AND CLUSTERING

For each selected path $P = (p_0, \dots, p_T)$, we compute local turning angle

$$\vartheta_t = \arccos\left(\frac{\langle p_t - p_{t-1}, p_{t+1} - p_t \rangle}{\|p_t - p_{t-1}\| \|p_{t+1} - p_t\| + \varepsilon}\right). \quad (7)$$

Candidate extraction follows the implementation constraints:

$$t \in \mathcal{C}(P) \iff \vartheta_t \geq \theta_{\text{turn}} \wedge \|p_t - p_{t_{\text{last}}}\|_2 \geq \delta_{\text{sep}}, \quad (8)$$

where t_{last} is the index of the last accepted turning point on the same path, sep stands for least acceptable distance between two adjacent turning points.

Each candidate (p_t, ϑ_t) has location $\text{location}(t) = (r_t, c_t)$ in the 2D grid cell. Define cross-path support count:

$$s_t = \sum_{k \in \mathcal{C}(P)} \mathbf{1}\{\text{location}(k) = \text{location}(t)\}, \quad (9)$$

then its adjusted priority is

$$\pi_t = \vartheta_t (1 + 0.25(s_t - 1)). \quad (10)$$

All candidates from all paths are pooled into $\mathcal{V}$. We then perform:

1. **Deduplication:** sort by (π_t, s_t) descending.
2. **Spatial clustering:** build a graph on deduplicated candidates with edge (i, j) iff $\|x_i - x_j\|_2 \leq \delta_{\text{clu}}$; each connected component $\mathcal{G}$ forms one cluster. Its center and score are

$$c_{\mathcal{G}} = \frac{\sum_{i \in \mathcal{G}} w_i x_i}{\sum_{i \in \mathcal{G}} w_i}, \quad w_i = \max(10^{-3}, \pi_i), \quad q_{\mathcal{G}} = \sum_{i \in \mathcal{G}} w_i. \quad (11)$$

Clusters are ranked by $q_{\mathcal{G}}$ (descending), and top M_{max} centers are kept as landmarks.

The resulting landmark set is $\mathcal{L} = \{\ell_1, \dots, \ell_M\}$, and final goal is appended as terminal landmark. In our default setup, $\theta_{\text{turn}} = 25^\circ$, $\delta_{\text{sep}} = 20\text{m}$, $\delta_{\text{clu}} = 25\text{m}$, and $M_{\text{max}} = 24$.

2.4 LANDMARK SELECTION WITH MEMORY PENALTY

At runtime, from the feasible candidate set $\mathcal{C}_t = \{\ell_1, \dots, \ell_K\}$ (unvisited landmarks with valid progress), we compute a multi-term score:

$$\text{Score}_i = \lambda_{\text{prog}} \Delta d_{\text{goal}}(\ell_i) - \lambda_{\text{mem}} \bar{M}(\ell_i) - \lambda_{\text{detour}} \Delta \text{Len}(\ell_i) + \lambda_{\text{safe}} \text{Clr}(\ell_i), \quad (12)$$

and convert scores into candidate weights

$$\alpha_i = \frac{\exp(\text{Score}_i / \tau_{\text{sel}})}{\sum_{j=1}^K \exp(\text{Score}_j / \tau_{\text{sel}})}, \quad \sum_{i=1}^K \alpha_i = 1. \quad (13)$$

λ_{prog} is the goal-progress weight, $\Delta d_{\text{goal}}(\ell_i)$ is the expected distance reduction to the final goal when considering candidate ℓ_i , λ_{mem} is the memory-penalty weight, and $\bar{M}(\ell_i)$ is local memory intensity around ℓ_i . λ_{detour} is the detour-penalty weight, $\Delta \text{Len}(\ell_i)$ is the extra path length induced by ℓ_i , λ_{safe} is the safety weight, and $\text{Clr}(\ell_i)$ is the obstacle-clearance margin near ℓ_i . τ_{sel} is the temperature for score-to-weight conversion.

2.5 GOAL-CONDITIONED PPO AND TRAJECTORY MEMORY UPDATE

Given candidate landmarks from Section 2.4, the low-level controller follows policy $\pi_\theta(a_t | o_t, \mathcal{C}_t, \alpha_t)$. Observation includes local obstacle cue, relative goal cue, and candidate landmark cues.

The total reward is

$$r_t = w_{\text{prog}} r_t^{\text{lm, max}} + w_{\text{event}} r_t^{\text{event}} - w_{\text{step}} - w_{\text{mem}} \hat{M}(p_t), \quad (14)$$

w_{prog} is the dense-progress reward weight, and $w_{\text{event}} r_t^{\text{event}}$ is event reward/penalty. w_{step} is the step cost coefficient and w_{mem} is the memory penalty weight. $\hat{M}(p_t)$ is the normalized memory value at the current position p_t . The max-aggregated landmark progress term is

$$r_t^{\text{lm, max}} = \max_{i \in \{1, \dots, K\}} [\alpha_i (\|p_{t-1} - \ell_i\| - \|p_t - \ell_i\|)]. \quad (15)$$

$r_t^{\text{lm, max}}$ implements winner-take-all credit assignment among candidates.

At decision points, candidate set $\mathcal{C}_t$ and weights α_t are refreshed. After each episode trajectory τ for initial i , external memory updates as

$$\mathcal{M}_i \leftarrow \text{clip}(\eta \mathcal{M}_i + \mathbf{1}_{\text{succ}} \text{Repulse}(\tau), 0, M_{\text{max}}), \quad (16)$$

$\mathcal{M}_i$ is the external memory map for initial condition i , decayed by η , which is the memory-retention factor. $\mathbf{1}_{\text{succ}}$ is the success indicator, activated update only for successful rollouts. $\text{Repulse}(\tau)$ is the trajectory-to-grid repulsion injection operator, and M_{max} is the memory cap.

3 EXPERIMENTS

3.1 EXPERIMENTAL SETUP

Environment and Data We use the provided 2D urban navigation environment constructed from point cloud obstacles. Evaluation follows the official protocol on the predefined start-goal set, and each start-goal pair requires multiple successful trajectories.

Table 2: Per-initial diversity scores (baseline vs. ours).

Initial ID	Baseline	Ours	Initial ID	Baseline	Ours
0	2644.67	2260.59	10	1079.39	1549.47
1	2384.69	2921.61	11	219.34	357.70
2	2091.64	1687.98	12	2714.04	2835.84
3	2682.97	3079.89	13	281.40	3286.81
4	917.92	683.39	14	837.72	1204.30
5	1130.83	1480.78	15	1121.20	3479.21
6	2953.20	1239.31	16	1013.62	1833.46
7	2413.48	3121.24	17	1934.18	3079.53
8	216.36	776.69	18	344.26	1211.05
9	494.24	2794.89	19	12452.98	8504.42

3.2 METRICS

We report average pairwise DTW distance after arc-length resampling (official metric).

For a start-goal pair with trajectory set $\mathcal{T} = \{\tau_1, \dots, \tau_n\}$, diversity is

$$\text{Div}(\mathcal{T}) = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} \text{DTW}(\tau_i, \tau_j), \quad (17)$$

and we report mean diversity across all initials.

3.3 BASELINES

We compare against one baseline only: the simple PPO implementation provided in `_instructor/`.

3.4 MAIN RESULTS

Table 1 summarizes quantitative results. Our method improves overall diversity from 1996.41 to 2369.41 (+18.68%). The per-initial median improves from 1126.01 to 2047.03 (+81.79%), and the minimum score improves from 216.36 to 357.70 (+65.33%). Although the maximum value decreases (12452.98 to 8504.42), this indicates reduced outlier dominance and a more balanced distribution across initials.

Table 1: Main comparison on the official evaluation protocol

Method	Overall DTW $\uparrow$	Min DTW $\uparrow$	Median DTW $\uparrow$
Baseline(PPO)	1996.41	216.36	1126.01
Ours	2369.41	357.70	2047.03
Relative Improvement	+18.68%	+65.33%	+81.79%

Detailed per-initial results are provided in Table 2.

3.5 ANALYSIS

Our method outperforms baseline on 15 out of 20 initials. The largest gains appear on initials 13, 15, and 9, where diversity increases by +3005.41, +2358.01, and +2300.65 respectively. These are typical mode-collapse cases for simple PPO, and the multi-path + memory mechanism substantially broadens route exploration.

The main regressions occur on initials 19, 6, and 2. In these cases, we observe that strong memory pressure can over-penalize effective corridors and push trajectories toward longer detours or unstable transitions, reducing pairwise DTW. This suggests that adaptive memory scheduling and per-scene tuning of repulsion strength remain important.



Figure 1: Comparison of trajectory distributions.

3.6 DISCUSSION

Compared with baseline PPO, explicit global-route diversification (dynamic-cost multi-path A* + clustered turning-point landmarks) is the key source of improvement. The memory penalty contributes to anti-collapse behavior by discouraging repeated corridor reuse, while landmark selection keeps local control feasible. The remaining failure cases motivate future work on adaptive memory decay and confidence-aware landmark switching.

4 CONCLUSION AND FUTURE WORK

We present a practical pipeline for diversified UAV trajectory collection that integrates dynamic-cost multi-path A*, turning-point landmark processing, and memory-conditioned PPO. On official metrics, our method outperforms the instructor simple PPO baseline by +18.68% in overall diversity and improves 15/20 initials. Future work will target stronger stability under hard initials and extension to full visual-control data collection for downstream VLA training.